

Game Theory

Section Id :	64065374344
Section Number :	15
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	6
Number of Questions to be attempted :	6
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653157578
Question Shuffling Allowed :	No

Question Number : 281 Question Id : 6406531031907 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY AND STRATEGY (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533489744.  YES

6406533489745.  NO

Sub-Section Number :	2
Sub-Section Id :	640653157579
Question Shuffling Allowed :	No

Question Id : 6406531031908 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (282 to 287)

Question Label : Comprehension

Consider the example of cascade discussed in the lecture but with a modification that now the urn is containing either 4 blue balls and 1 red ball or 4 red balls and 1 blue ball with equal probability. That simply means that the urn is equally likely to be blue-majority urn or red-majority urn. We assume everyone hears what the previous individuals report and get their own private signal by randomly drawing one ball from the urn.

Based on the above data answer the given subquestions.

Sub questions

Question Number : 282 Question Id : 6406531031909 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Suppose one ball is picked from the blue majority urn. What is the probability that the ball is blue?

Options :

6406533489746. ✖ 1/4

6406533489747. ✖ 3/4

6406533489748. ✖ 1/5

6406533489749. ✔ 4/5

Question Number : 283 Question Id : 6406531031910 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the probability that the urn is blue-majority if the first student draws a red ball?

Options :

6406533489750. ✔ 1/5

6406533489751. ✖ 4/5

6406533489752. ✖ 3/4

6406533489753. ✖ 1/4

Question Number : 284 Question Id : 6406531031911 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Blue, Blue, Red).

What is the probability of getting this sequence if the urn is a blue-majority urn?

Options :

6406533489754. ✖ 3/64

6406533489755. ✖ 4/125

6406533489756. ✔ 16/125

6406533489757. ✖ 9/125

Question Number : 285 Question Id : 6406531031912 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Blue, Blue, Red).

What is the probability of getting this sequence if the urn is a red-majority urn?

Options :

6406533489758. ✖ 3/64

6406533489759. ✖ 9/125

6406533489760. ✖ 16/125

6406533489761. ✔ 4/125

Question Number : 286 Question Id : 6406531031913 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Blue, Blue, Red).

What is the total probability of getting this sequence?

Options :

6406533489762. ✔ 2/25

6406533489763. ✖ 9/25

6406533489764. ✖ 6/125

6406533489765. ✖ 3/125

Question Number : 287 Question Id : 6406531031914 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Blue, Blue, Red).

What is the probability of urn being red-majority in this scenario?

Options :

6406533489766. ✖ 3/25

6406533489767. ✔ 1/5

6406533489768. ✖ 4/5

6406533489769. ✖ 9/25

Question Id : 6406531031929 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (288 to 293)

Question Label : Comprehension

Each one of two political parties can choose to buy time on commercial radio shows to broadcast negative ad campaigns against their rival. These choices are made simultaneously. Due to government regulation, it is forbidden to buy more than 2 hours of negative campaign time so that each party cannot choose an amount of negative campaigning above 2 hours. Given a pair of choices (a_1, a_2) , the payoff of party i is given by the following function: $v_i(a_1, a_2) = a_i - 2a_j + a_1a_2 - (a_i)^2$. Write the normal form representation of this game and answer the given sub-questions.

Sub questions

Question Number : 288 Question Id : 6406531031930 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If player 1 chooses 1.5 hour as his action, the best response of player 2 will be

Options :

6406533489794. ✖ 1.5

6406533489795. ✖ 1

6406533489796. ✔ 1.25

6406533489797. ✖ 0

Question Number : 289 Question Id : 6406531031931 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How much negative campaign time each player will choose under a symmetric and unique Nash equilibrium?

Options :

6406533489798. ✖ 1.5

6406533489799. ✔ 1

6406533489800. ✖ 1.25

6406533489801. ✖ 0

Question Number : 290 Question Id : 6406531031932 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Equilibrium payoff of each player will be

Options :

6406533489802. ✖ -1.5

6406533489803. ✖ 1

6406533489804. ✔ -1

6406533489805. ✖ 0

Question Number : 291 Question Id : 6406531031933 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now assume that this game is repeated infinitely. If the parties cooperate and sign a binding agreement on how much to campaign, they choose $a_1 = a_2 = 0$.

What will be the payoff of each player in each stage game if they stick to this agreement

Options :

6406533489806. ✖ -1.5

6406533489807. ✖ 1

6406533489808. ✖ -1

6406533489809. ✔ 0

Question Number : 292 Question Id : 6406531031934 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Now assume that this game is repeated infinitely. If the parties cooperate and sign a binding agreement on how much to campaign, they choose $a_1 = a_2 = 0$.

What should be the level of campaign a player should choose if wants to deviate from the agreement to get the payoff of $\frac{1}{4}$

Options :

6406533489810. ✖ 1

6406533489811. ✔ $\frac{1}{2}$

6406533489812. ✖ $\frac{1}{3}$

6406533489813. ✖ $\frac{1}{4}$

Question Number : 293 Question Id : 6406531031935 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Now assume that this game is repeated infinitely. If the parties cooperate and sign a binding agreement on how much to campaign, they choose $a_1 = a_2 = 0$.

For which discount factors $\delta \in (0, 1)$ can this agreement not be supported as a subgame perfect equilibrium of the infinitely repeated game?

Options :

6406533489814. ✖ $\delta \geq \frac{1}{5}$

6406533489815. ✔ $\delta \leq \frac{1}{5}$

6406533489816. ✖ $\delta \geq \frac{1}{3}$

6406533489817. ✖ $\delta \leq \frac{1}{3}$

Sub-Section Number : 3
 Sub-Section Id : 640653157580
 Question Shuffling Allowed : No

Question Id : 6406531031915 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
 Question Numbers : (294 to 297)

Question Label : Comprehension

Given the following preference structure, find a stable matching system using the Gale–Shapley algorithm, when the women propose:

Women: A, B, C.				Men: a, b, c, d.			
	A	B	C		A	B	C
a	1	1	0	a	0	0	1
b	2	0	1	b	3	1	2
c	3	2	3	c	2	1	0
d	4	0	2	d	2	3	1

Based on the above data answer the given subquestions.

Sub questions

Question Number : 294 Question Id : 6406531031916 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Mr. a will be paired with

Options :

- 6406533489770. ✖ A
- 6406533489771. ✖ B
- 6406533489772. ✖ C
- 6406533489773. ✔ None

Question Number : 295 Question Id : 6406531031917 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Mr. b will be paired with

Options :

6406533489774. ✖ A

6406533489775. ✖ B

6406533489776. ✔ C

6406533489777. ✖ None

Question Number : 296 Question Id : 6406531031918 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Mr. c will be paired with

Options :

6406533489778. ✖ A

6406533489779. ✔ B

6406533489780. ✖ C

6406533489781. ✖ None

Question Number : 297 Question Id : 6406531031919 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Mr. d will be paired with

Options :

6406533489782. ✔ A

6406533489783. ✖ B

6406533489784. ✖ C

6406533489785. ✖ None

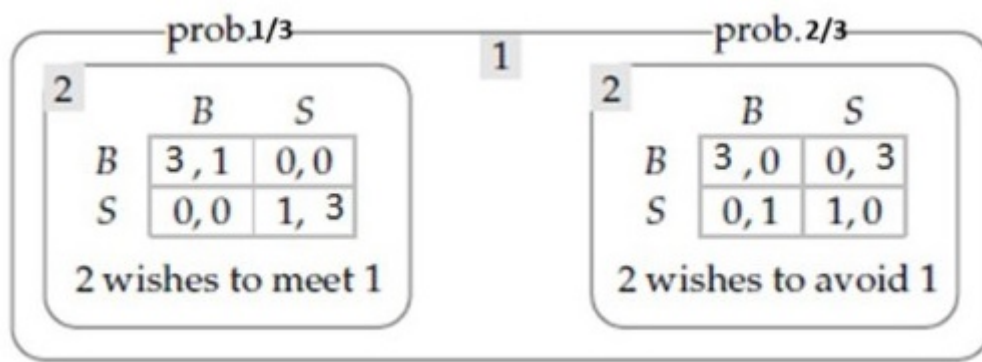
Question Id : 6406531031920 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (298 to 305)

Question Label : Comprehension

(Bayesian Game) Consider a variant of the situation modelled by *Battle of Sexes (BoS)* (as discussed in lectures) in which player 1 is unsure whether player 2 prefers to go out with her or prefers to avoid her, whereas player 2, as before, knows player 1's preferences. Specifically, suppose player 1 thinks that with probability $1/3$ player 2 wants to go out with her, and with probability $2/3$ player 2 wants to avoid her. The situation is depicted in the following matrix of a static game with incomplete information:



With the help of this game matrix, we may come up with the following table depicting expected payoff of player 1 corresponding to different actions of both types of player 2.

		Player 2 (both types)			
		(B, B)	(B, S)	(S, B)	(S, S)
Player 1	B	a	b	c	d
	S	p	q	r	t

Based on the above data answer the given subquestions.

Sub questions

Question Number : 298 Question Id : 6406531031921 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of a = _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 299 Question Id : 6406531031922 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of b = _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 300 Question Id : 6406531031923 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $c =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 301 Question Id : 6406531031924 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $d =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 302 Question Id : 6406531031925 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $p =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 303 Question Id : 6406531031926 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $q =$ _____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.66 to 0.68

Question Number : 304 **Question Id :** 6406531031927 **Question Type :** SA
Correct Marks : 0.5

Question Label : Short Answer Question
Find out the value of $r =$ _____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.33 to 0.34

Question Number : 305 **Question Id :** 6406531031928 **Question Type :** SA
Correct Marks : 0.5

Question Label : Short Answer Question
Find out the value of $t =$ _____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Sub-Section Number : 4
Sub-Section Id : 640653157581
Question Shuffling Allowed : Yes

Question Number : 306 **Question Id :** 6406531031936 **Question Type :** MCQ
Correct Marks : 1

Question Label : Multiple Choice Question
Consider two statements about Borda count:
I: Borda Count (ignoring ties) always produces a complete, transitive ranking for a set of alternatives.

II: Outcome in Borda count does not depend on the presence of alternatives that seem “irrelevant”.

Choose the correct alternative about these statements:

Options :

6406533489818. ✔ Only I is true

6406533489819. ✖ Only II is true

6406533489820. ✖ None is true

6406533489821. ✖ Both are true